Jonathan Hernandez Rodriguez

File and Directory Management in Linux

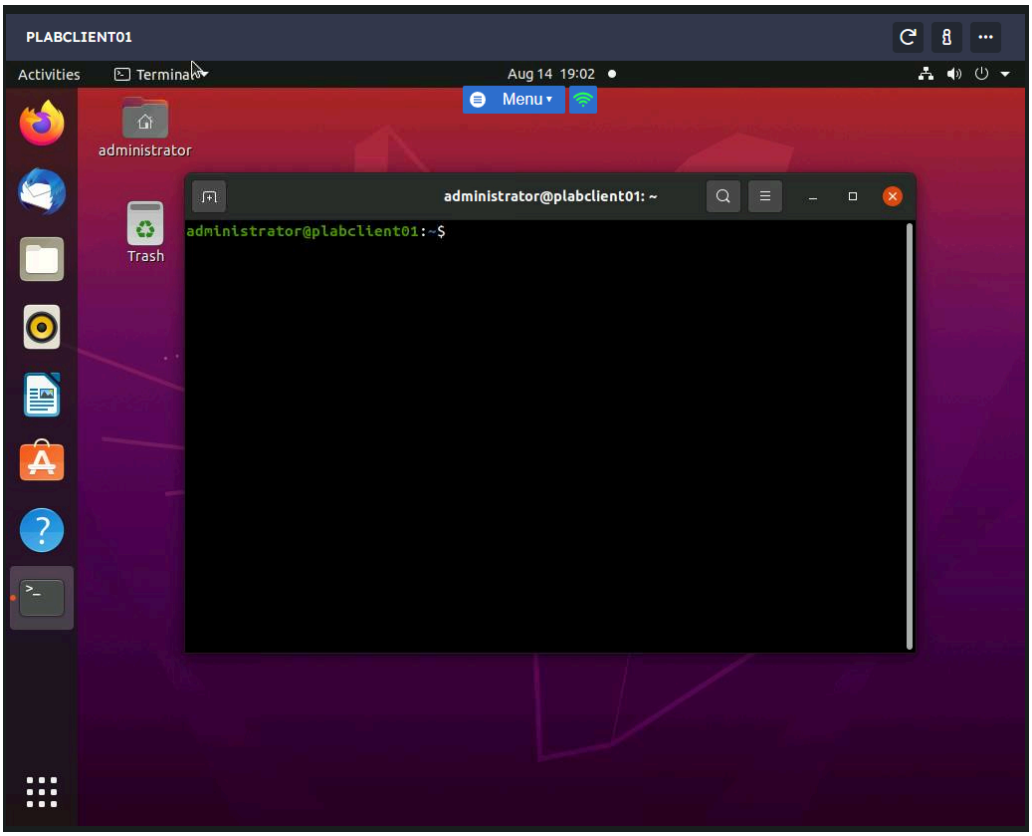
Exercise 1 - Using the Linux Terminal

Task 1: Launch the GNOME Terminal

In this task, I will open the GNOME Terminal and get acquainted with the command-line interface.

First, I'll connect to PLABCLIENT01. Once the login screen appears, I'll select the administrator user. After entering the password (Passw0rd), I'll press Enter to log in.

Once I'm logged in, I'll locate the Show Applications icon on the left pane and click on it to view all installed applications. Then, I'll scroll down to find Terminal, right-click on it, and select Add to Favorites to make it easily accessible. The Terminal icon will now appear on the left pane. I'll click on it to open the Terminal window and keep this window open as I move on to the next task.



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Task 2: Customize the Terminal

In this task, I will customize the appearance of the GNOME Terminal to suit my preferences.

With the Terminal window open, I'll notice three icons at the top: Search (a magnifying glass), Menu (three horizontal lines), and Tabs (a plus sign). I'll begin by clicking on the Menu icon and selecting Preferences from the drop-down menu.

I'll find the Theme variant drop-down list in the Preferences- General window. I'll click on it and select Light to switch to a lighter theme. Next, on the left pane, I'll click on Unnamed to access the profile settings. I'll click the drop-down menu next to Unnamed and select Clone.

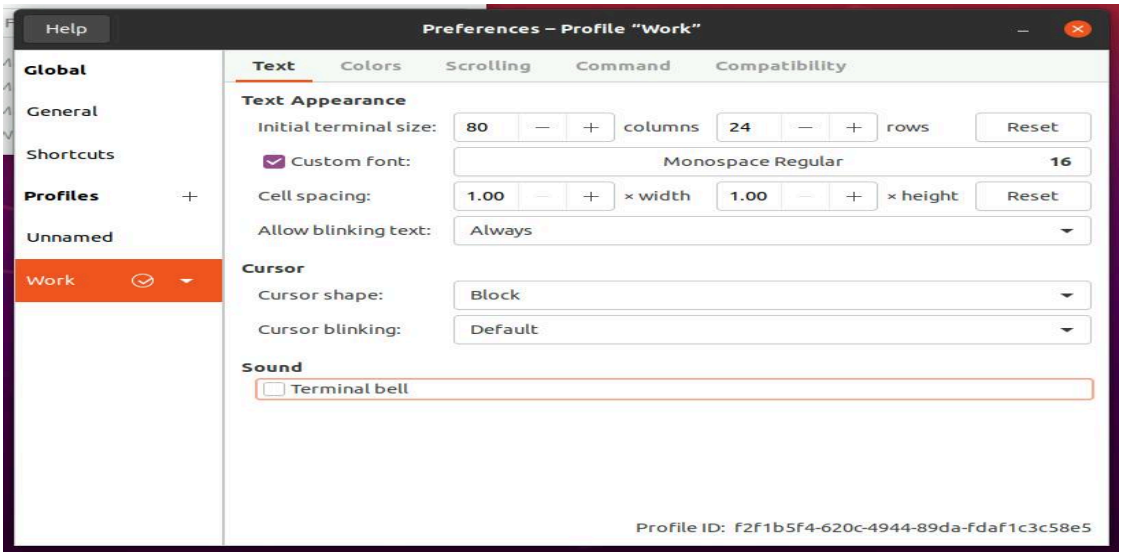
In the Clone Profile dialog box, I'll enter Work as the name for my new profile and click Clone. This creates a new profile where I can apply my custom settings.

Now, I'll enable the Custom font option in the Preferences - Profile "Work" pane by ticking its checkbox. I'll click on the current font (Monospace Regular) to open the font selection dialog. In the Choose A Terminal Font dialog box, I'll increase the font size to 16 by clicking the plus sign next to the size field, then click Select.

Back in the Preferences - Profile "Work" window, I'll untick the Terminal bell checkbox to disable the terminal bell sound. I'll notice that the Unnamed profile is still marked as the default profile. To change this, I'll click the drop-down menu next to the Work profile and select Set as default.

After completing these changes, I'll close the Preferences - Profile "Work" window and the Terminal window.

Finally, I'll click the Terminal icon in the left (favorites) pane to relaunch it. I'll notice that the text now appears with a font size of 16. If time permits, I can further customize the terminal to suit my preferences.



Exercise 2 - Navigate the Linux File System

Task 1: Use Basic Commands to Navigate the Linux File System

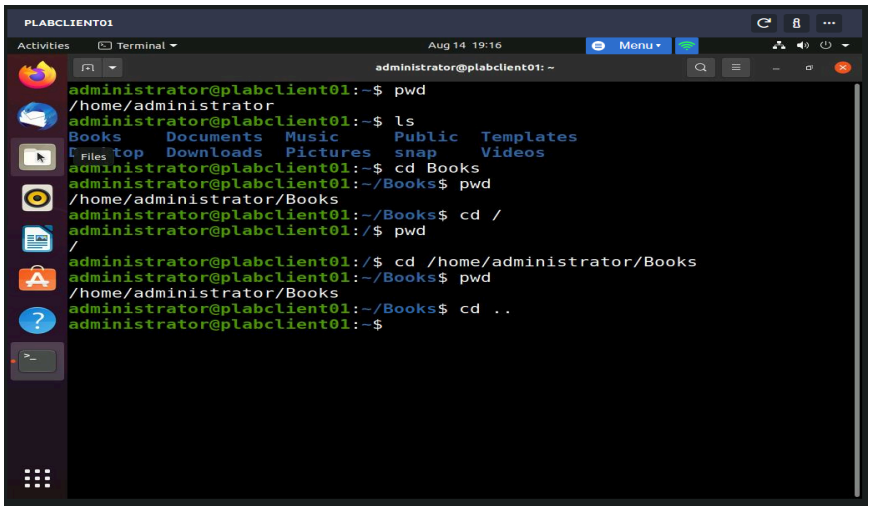
First, I'll connect to PLABCLIENT01, where I'll already have the Terminal window open. The command prompt, administrator@plabclient01:~$, indicates that I am logged in as the administrator on the plabclient01 system, and I'm currently in the home directory (~). If I were logged in as the root user, the prompt would be root@plabclient01:/home/administrator#, where the # symbol indicates root privileges. However, since root login is typically disabled by default in Ubuntu, I'm operating under the standard user prompt.

To begin, I'll type the command pwd and press Enter. This command displays the present working directory, which confirms my current location within the file system. In Linux, the root of the file system is denoted by /, with /home as a sub-directory and /home/administrator as my specific home directory.

Next, I'll use the ls command to list all the subdirectories within my current directory.
I'll then navigate into a specific subdirectory by typing cd Books and pressing Enter. The cd command allows me to change directories, and by entering Books, I move into that subdirectory. notice that the prompt now reflects this change, showing that I am no longer in the home directory.

To confirm my location, I'll type pwd again, which should display the path /home/administrator/Books. To return to the root directory, I'll type cd / and then pwd to verify that I've moved to the root directory. Afterward, I'll navigate back to the Books directory using cd /home/administrator/Books and confirm my location with pwd.

Finally, I'll move up one directory level by typing cd .. and pressing Enter, which takes me back to the home directory. To clear the Terminal screen of all previous commands and outputs, I'll type clear.



Task 2: Create Directories in the Linux File System

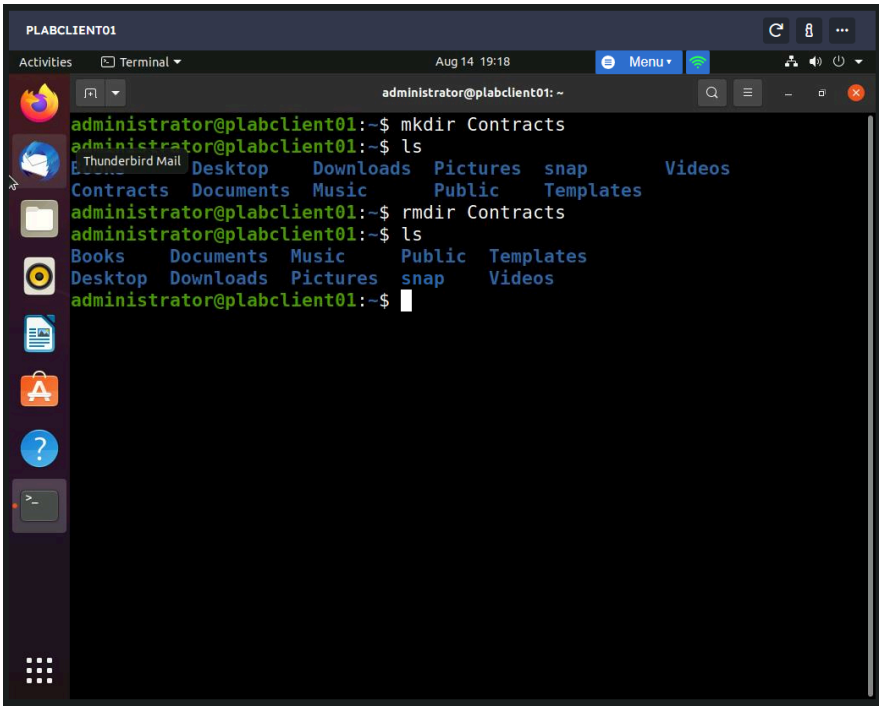
In this task, I will create and remove a directory within the Linux file system.

Starting from the home directory in the Terminal window, I'll create a new directory called Contracts by typing mkdir Contracts and pressing Enter. The mkdir command is used for making new directories. To confirm that the Contracts directory has been successfully created, I'll use the ls command to list the contents of my current directory.

Once I see that the Contracts directory has been created, I'll remove it by typing rmdir Contracts and pressing Enter. The rmdir command is used to delete empty directories. To verify that the directory has been removed, I'll use ls again to check the current directory contents. The Contracts directory should no longer be listed.

Finally, I'll clear the Terminal window again by typing clear before proceeding to the next exercise. This keeps my workspace tidy and focused.

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Exercise 2 - Create Files in the Linux File System

Task 1: Update and Upgrade the Ubuntu System

First, I will update and upgrade the Ubuntu system to ensure it is running the latest software. I'll begin by connecting to PLABCLIENT01 and opening the Terminal window.

In the Terminal, I will type sudo apt-get update and press Enter. The system will prompt me for the password, so I'll enter Passw0rd. After that, I will run the sudo apt-get upgrade command to upgrade the system. These commands are essential for keeping the system secure and up-to-date with the latest software patches and updates.

When the prompt asks, "Do you want to continue?" I will type Y and press Enter to confirm that I want to proceed with the upgrade.

Task 2: Navigate to the Documents Directory

Now that the system is updated, I will navigate to the Documents directory to start creating and managing files.

To do this, I will type cd Documents in the Terminal and press Enter. This command will take me directly to the Documents directory. To see what's in the directory, I will use the ls command, which lists all the contents of the current directory.

This step sets the stage for file creation and manipulation within the Documents directory.

Task 3: Create and Edit Files Using Nano Editor

With the Documents directory open, I'll create and edit some files using the nano text editor.

First, I'll create a new file called example1.txt by typing touch example1.txt and pressing Enter. The touch command is a quick way to create an empty file. After running ls, I should see example1.txt listed in the directory.

Next, I'll create and edit another file using the nano text editor. I'll type nano example2.txt and press Enter. This will open the nano editor. In the file, I will write, "I have created this file using nano. I will use the shortcuts Ctrl+O and Ctrl+X to save the file and close nano." After typing, I'll save the file by pressing Ctrl+O and then pressing Enter. To exit nano, I'll press Ctrl+X.

Task 4: Install Vim and Create a File Using Vim Editor

In this task, I'll install the vim editor, which is not installed by default on Ubuntu, and then use it to create a file.

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To install vim, I'll type sudo apt install vim in the Terminal and press Enter. I will enter the password Passw0rd when prompted and confirm the installation by typing Y at the prompt.

Once vim is installed, I'll create a new file by typing vim example3.txt and pressing Enter. In vim, I'll press i to enter Insert mode and type, "This is an example of a file created using the vim editor." To save and exit vim, I'll press Esc, type :wq, and press Enter.

Finally, to view the contents of example3.txt, I'll type cat example3.txt and press Enter. The cat command will display the contents of the file in the Terminal.

Before moving on to the next task, I will clear the Terminal by typing clear. This will keep the workspace organized for any further activities.

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Exercise 3 - Move Files and Create Links

Task 1: Move and View Files Using Linux Commands

To begin, I make sure I'm connected to PLABCLIENT01 with the Terminal window open. I navigate to the Books directory by entering cd ~/Books in the Terminal. Once there, I list the contents of the directory using the ls command to see what files are available.

Next, I create a copy of the file dracula.txt and rename it to Dracula.txt within the same directory. This is done with the command cp dracula.txt Dracula.txt. After that, I need to copy another file, frankenstein.txt, to the Documents directory and rename it to Frankenstein.txt. I achieve this by running cp frankenstein.txt ~/Documents/Frankenstein.txt, and then I navigate to the Documents directory using cd ~/Documents/ and list the contents with ls to confirm the file has been copied correctly.

Task 2: Remove and Rename Files

To continue, I return to the Books directory by typing cd ~/Books in the Terminal. My next task is to remove the file dracula.txt from the directory. I do this using the rm dracula.txt command, and I confirm its removal by listing the directory contents with ls.

Afterward, I rename the file Dracula.txt back to dracula.txt. This is accomplished by using the mv Dracula.txt dracula.txt command. Running ls again allows me to verify that the file has been successfully renamed.

Task 3: Create a Hard Link

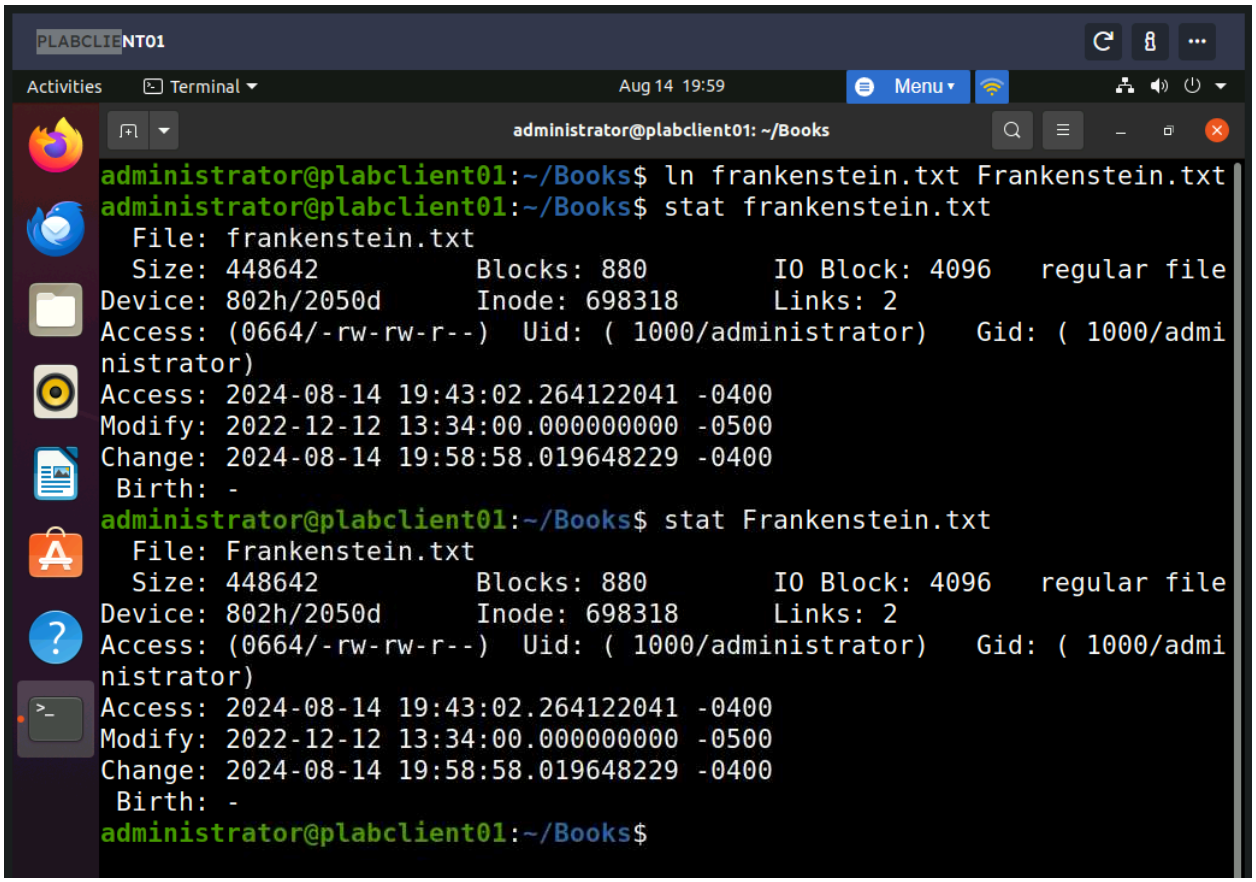
Now, I proceed to create a hard link, which is essentially an exact copy of the file that shares the same data blocks. To do this, I use the command ln dracula.txt dracula_hardlink.txt. This command creates a hard link of dracula.txt named dracula_hardlink.txt in the same directory.

I then list the contents of the Books directory with the ls -l command to confirm the creation of the hard link. The detailed listing shows the hard link count, ensuring that the hard link has been established correctly.

Task 4: Create a Soft Link

Finally, I create a soft link, which acts as a reference or shortcut to the original file rather than a complete copy. I use the command ln -s dracula.txt dracula_softlink.txt to create a soft link named dracula_softlink.txt pointing to dracula.txt within the same directory.

To verify that the soft link has been created, I list the directory contents again with ls -l. The soft link is indicated by the -> symbol, showing the link between dracula_softlink.txt and dracula.txt.



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Exercise 4 - Controlling Access to Directory and Files with Standard Permissions

Task 1: Create Files and View Permissions

To begin, I connected to the PLABCLIENT01 system and logged in as the administrator using the password "Passw0rd." Once logged in, I accessed the Terminal by clicking the Show Applications icon on the bottom left corner, scrolling down, and selecting Terminal.

In the Terminal, I navigated to the Documents folder by entering the command cd Documents/ and pressing Enter. Then, I created three files named textfile1.txt, textfile2.txt, and textfile3.txt using the touch command for each file. After executing these commands, I verified the creation of the files by listing the contents of the Documents folder with the ls command.

Next, I navigated to the /etc directory using the command cd /etc and pressed Enter. The /etc directory is crucial as it contains all the Linux system configuration files. To view the contents of the /etc/passwd file, I used the cat passwd command. This file contains user account information, and I was also able to view the file permissions.

When I attempted to view the contents of the /etc/shadow file using the cat shadow command, I received a "Permission denied" error. This happened because only root users or users belonging to the shadow group have access to this sensitive file, which stores encrypted user passwords.

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Task 2: Modify File Permissions

In this additional task, I modified the permissions of the files created in the previous task to better understand how Linux permissions work.

First, I ensured I was still in the Documents directory by typing cd ~/Documents and pressing Enter. Then, I checked the current permissions of the files textfile1.txt, textfile2.txt, and textfile3.txt by using the ls -l command. This command lists files along with detailed information, including their permissions.

Next, I changed the permissions of textfile1.txt to make it read-only for the owner, group, and others by typing chmod 444 textfile1.txt and pressing Enter. The chmod command modifies file permissions, and the number 444 indicates read-only access for all users.

For textfile2.txt, I changed the permissions to allow the owner to read and write, but only read access for the group and others. I achieved this by entering chmod 644 textfile2.txt and pressing Enter.

Finally, I set textfile3.txt to be executable by the owner, while the group and others had no permissions. This was done with the command chmod 700 textfile3.txt.

After modifying the permissions, I verified the changes by listing the files again using the ls -l command. The new permissions were displayed, confirming that the modifications were successful.

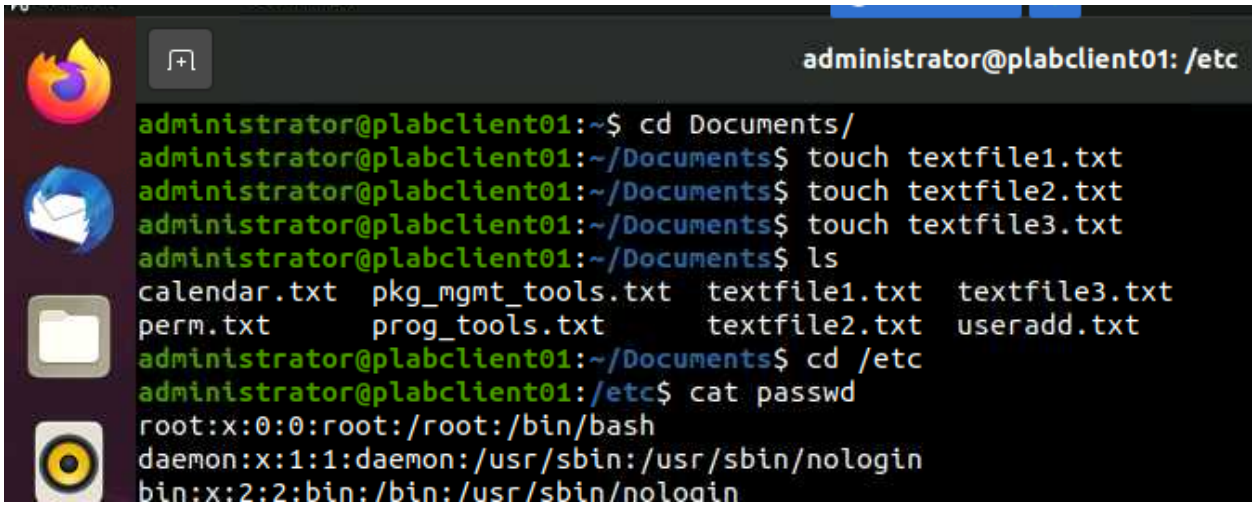
Task 3: Create Files and View Permissions

I started by connecting to the PLABCLIENT01 device and logged in using the administrator credentials. Once logged in, I opened the Terminal by clicking on the "Show Applications" icon at the bottom left corner and selecting Terminal from the menu.

In the Terminal, I navigated to the Documents directory by typing cd Documents/ and pressing Enter. Then, I created three text files named textfile1.txt, textfile2.txt, and textfile3.txt using the touch command. After creating these files, I confirmed their creation by listing the contents of the Documents directory using the ls command.

Next, I navigated to the /etc directory to explore system configuration files. I used the cat passwd command to view the contents and permissions of the /etc/passwd file. When I tried to view the /etc/shadow file using the cat shadow command, I received a "Permission denied" error, as only root users or users in the shadow group have access to this file.

Afterward, I viewed the permissions of the shadow file using the ls -l shadow command, which provided detailed information about the file, including the owner, group, and permission settings. To clear the Terminal window for further tasks, I used the clear command.



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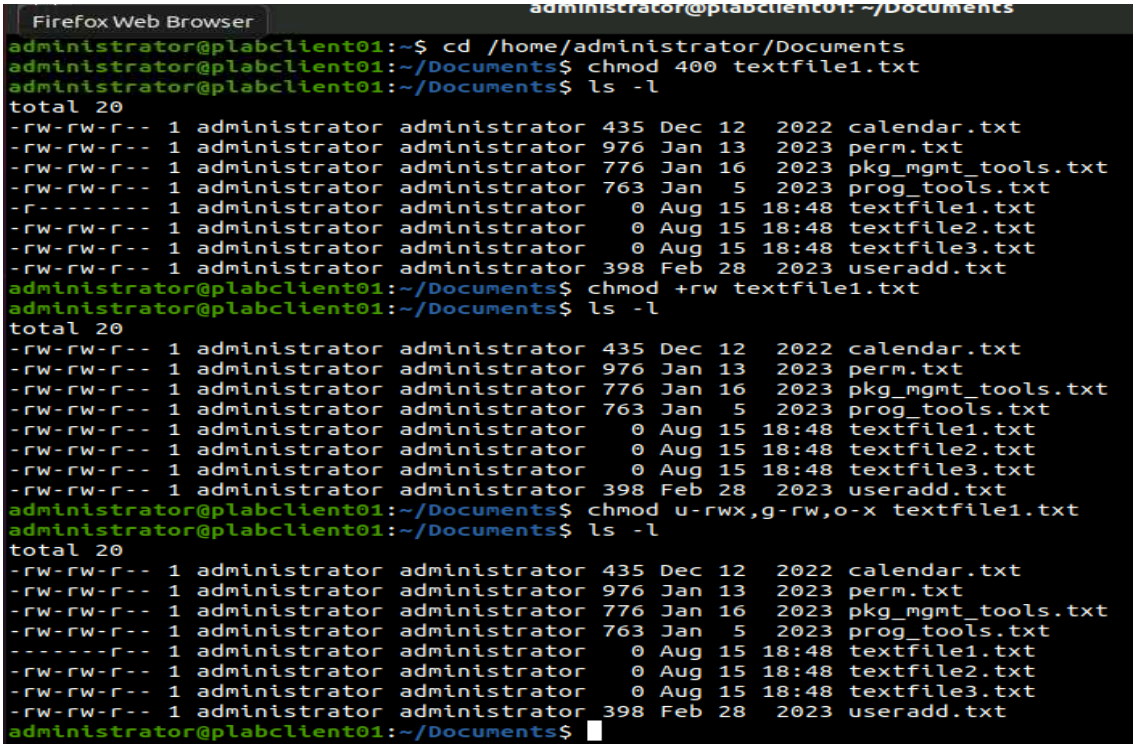
Task 4: Modify File Permissions

Continuing from the previous task, I focused on modifying the permissions of the files I created in the Documents directory. I started by navigating back to the Documents directory using cd /home/administrator/Documents and then changed the permissions of textfile1.txt to read-only for the owner, group, and others by using the chmod 400 textfile1.txt command.

To verify the change, I listed the files again with the ls -l command and observed that textfile1.txt had read-only permissions. I then modified the permissions again, this time granting read and write access to the owner, group, and others using the chmod +rw textfile1.txt command.

I removed all permissions for the owner of textfile1.txt while giving read access to all other users using chmod u-rwx,g-rw,o-x textfile1.txt. After checking the permissions with ls -l, I made an additional change by removing the write permission for the group using chmod g-w textfile1.txt.

Next, I navigated to the default home directory using the cd command and viewed the .profile file with the less ./.profile command. This command displayed the default permissions as defined by the Linux system, where I noticed the umask 022 setting, indicating a default permission of 755 for newly created files and directories.


Firefox Web Browser
administrator@plabclient01: ~/Documents
administrator@plabclient01:~$ cd /home/administrator/Documents
administrator@plabclient01:~/Documents$ chmod 400 textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 2022 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 2023 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 2023 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 2023 prog_tools.txt
-r-------- 1 administrator administrator 0 Aug 15 18:48 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 2023 useradd.txt
administrator@plabclient01:~/Documents$ chmod +rw textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 2022 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 2023 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 2023 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 2023 prog_tools.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 2023 useradd.txt
administrator@plabclient01:~/Documents$ chmod u-rwx,g-rw,o-x textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 2022 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 2023 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 2023 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 2023 prog_tools.txt
---------- 1 administrator administrator 0 Aug 15 18:48 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 Aug 15 18:48 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 2023 useradd.txt
administrator@plabclient01:~/Documents$

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Task 5: Control Ownership and Group Permissions

To explore ownership and group control, I created a new user named jdoe using the sudo useradd jdoe command. After setting the password, I navigated back to the Documents directory and changed the ownership of textfile2.txt to the newly created user jdoe using the sudo chown jdoe textfile2.txt command. I verified the change with the ls -l command, where jdoe was now listed as the owner.

Next, I created a new group called Marketing using the sudo groupadd Marketing command. I then changed the group ownership of textfile2.txt to Marketing using the sudo chgrp Marketing textfile2.txt command. To further practice, I used the sudo chown :Marketing textfile3.txt command to change the group ownership of textfile3.txt to Marketing.

Finally, I reset the ownership and group permissions of all files in the Documents directory to the default administrator settings using sudo chown -R administrator:administrator ./* and confirmed the changes with ls -l.

```
administrator@plabclient01:~/Documents$ cd
administrator@plabclient01:~$ less ./.profile
administrator@plabclient01:~$ id
uid=1000(administrator) gid=1000(administrator) groups=1000(administrator),4(ad
dip),46(plugdev),120(lpadmin),132(lxd),133(sambashare)
a   strator@plabclient01:~$ sudo useradd jdoe
[sudo] password for administrator:
administrator@plabclient01:~$ cd Documents.
bash: cd: Documents.: No such file or directory
administrator@plabclient01:~$ cd Documents/
administrator@plabclient01:~/Documents$ sudo chown jdoe textfile2.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12  2022 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13  2023 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16  2023 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan  5  2023 prog_tools.txt
-------r-- 1 administrator administrator   0 Aug 15 18:48 textfile1.txt
-rw-rw-r-- 1 jdoe          administrator   0 Aug 15 18:48 textfile2.txt
-rw-rw-r-- 1 administrator administrator   0 Aug 15 18:48 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28  2023 useradd.txt
administrator@plabclient01:~/Documents$
```

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Exercise 5 - Controlling Access to Directory and Files with Advanced Permissions

Task 1 - Verify ACL Support and Set Up Initial User Permissions

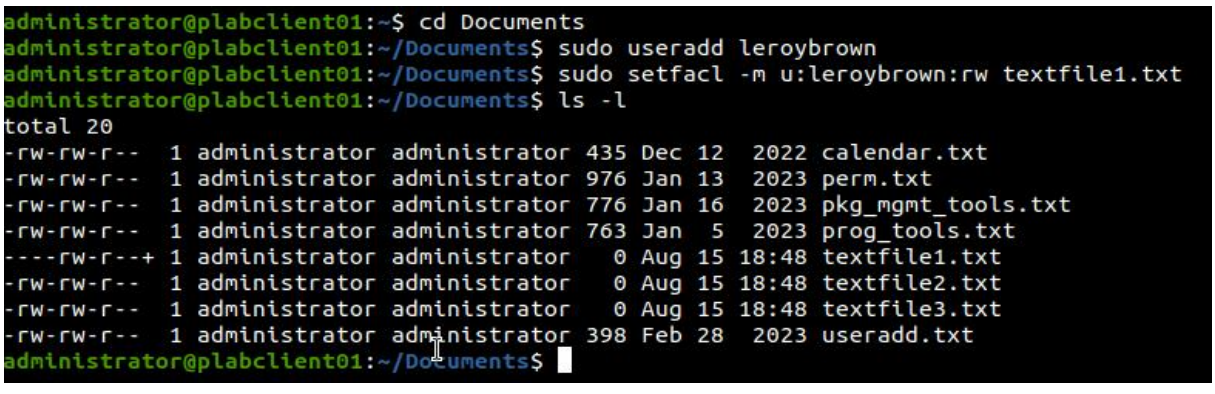
To begin, I ensured I was connected to the system and had the Terminal window open. My first objective was to verify if the file system supported Access Control Lists (ACLs). I achieved this by running the command sudo tune2fs -l /dev/sda2, which confirmed that the ACL option was enabled, allowing the use of FACLs.

Next, I navigated to the Documents directory by typing cd Documents. I then created a new user named leroybrown using the command sudo useradd leroybrown. With the user created, I proceeded to grant read and write permissions to leroybrown on the file textfile1.txt by executing sudo setfacl -m u:leroybrown:rw textfile1.txt. To verify the applied permissions, I used the command sudo getfacl textfile1.txt, which displayed the current ACL settings for the file.

Task 2 - Modify and Remove ACL Entries

With the initial ACL setup complete, I moved on to modifying and removing ACL entries. I decided to remove the ACL entry for leroybrown from textfile1.txt by executing sudo setfacl -x u:leroybrown textfile1.txt. I then confirmed the removal of permissions by running sudo getfacl textfile1.txt again, ensuring that leroybrown no longer had access to the file.

To continue with the modifications, I granted read and write permissions to the Marketing group for textfile1.txt. I did this by executing the command sudo setfacl -m g:Marketing:rw textfile1.txt, and then I verified the updated ACL settings using sudo getfacl textfile1.txt, confirming that the Marketing group had the correct permissions.



Task 3 - Create a Directory and Apply User-Specific ACLs

The next step involved creating a new directory named Reports. I created this directory by typing mkdir Reports. Afterward, I applied read and write permissions specifically for the user leroybrown within the Reports directory using the command sudo setfacl -m u:leroybrown:rw Reports.

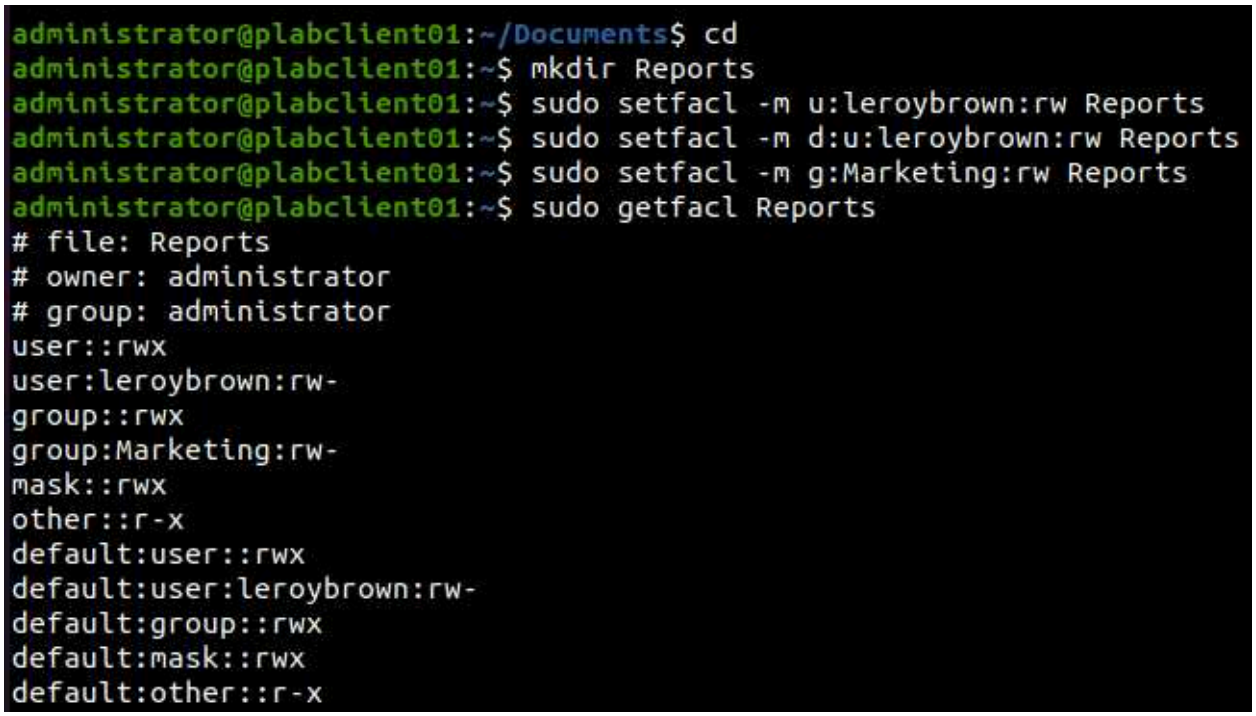
I then focused on ensuring that these permissions would be inherited by any future files or subdirectories created within Reports. To achieve this, I set up permission inheritance by running the command sudo setfacl -m d:u:leroybrown:rw Reports, making sure that leroybrown would have consistent access across all contents in the Reports directory.



Task 4 - Set Up Group-Specific ACLs and Permission Inheritance

Building on the previous task, I configured ACLs for the Marketing group within the Reports directory. I did this by applying read and write permissions for the group using the command sudo setfacl -m g:Marketing:rw Reports.

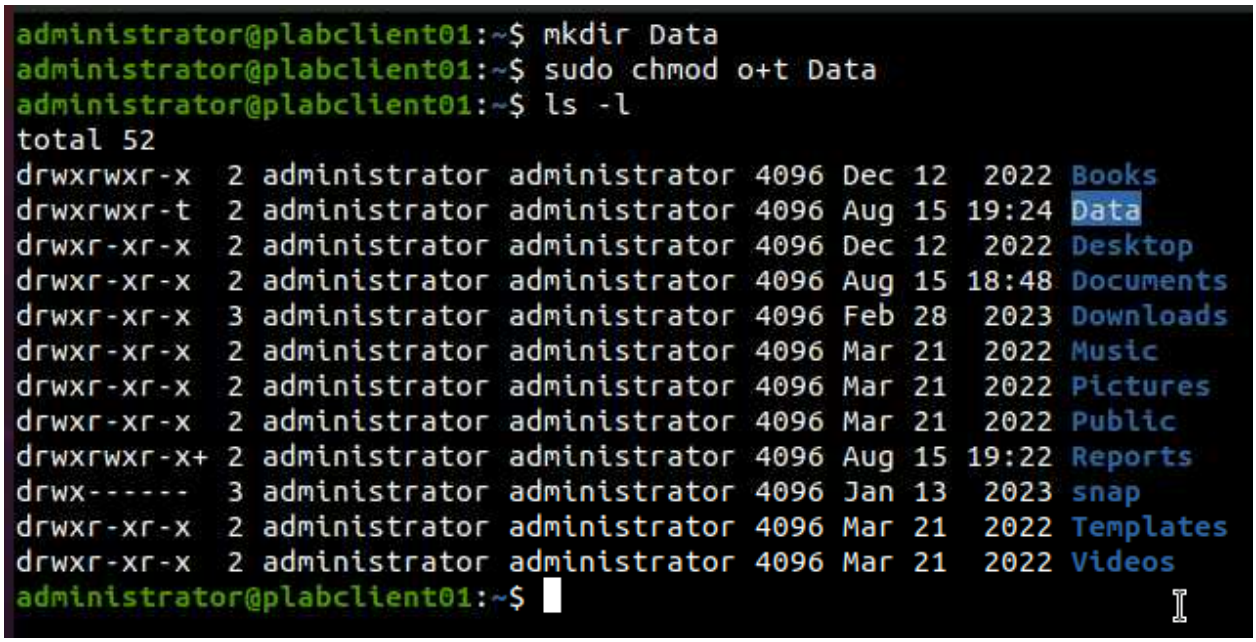
To confirm that both the user leroybrown and the Marketing group had the correct permissions, I ran the command sudo getfacl Reports. This allowed me to verify that permission inheritance was properly configured for both the user and the group within the Reports directory.



Task 5 - Implement and Verify Sticky Bit Permissions

In the final task, I focused on implementing sticky bit permissions to control access within a shared directory. First, I created a directory named Data by using the command mkdir Data. To restrict file modifications within this directory, I applied the sticky bit by running sudo chmod o+t Data. This ensured that only the file owners and root users could modify or delete files within the Data directory.

To verify that the sticky bit was correctly set, I checked the directory permissions by typing ls -l. The presence of the t at the end of the permission string indicated that the sticky bit was active, successfully limiting unauthorized modifications to the directory's contents.



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Exercise 6 - Create and Configure User Accounts in Linux

Task 1 - Create a User Account

To start, I connected to PLABCLIENT01 and logged in as the administrator using the password Passw0rd. Once logged in, I navigated to the Show Applications icon located at the bottom left corner of the screen. From there, I selected Terminal to open a command line interface.

In the Terminal, I created a new user account named janedoe by executing the command: sudo useradd janedoe -c "Jane Doe" -s /bin/sh -d /home/janedoe -e 2026/12/31. This command set up the user with a specific display name, default shell, home directory, and an account expiry date. After entering the administrator password to authorize the command, I ran sudo useradd -D to display the default values for new users.

Next, I used the command sudo useradd -m janedoe to create the home directory for janedoe with the default settings. Then, I set a password for the new account using the sudo passwd janedoe command, entering Passw0rd when prompted to set and confirm the password.

After setting up the account, I logged out from the administrator session and logged in as Jane Doe using the new credentials to ensure everything was set up correctly. I then logged back out and returned to the administrator account.

Task 2 - Modify the User Account

After successfully creating the user account, I proceeded to modify it. I opened the Terminal and renamed the user account from janedoe to joedoe by executing the command sudo usermod -l joedoe janedoe. To confirm the change, I used sudo less /etc/passwd and verified that the account was listed under the new name.

Next, I locked the joedoe account to prevent it from being used for login by running sudo usermod -L joedoe. To unlock the account, I used sudo usermod -U joedoe. Finally, I changed the shell for the joedoe account to make it a service account, which prevents interactive logins, by executing sudo chsh -s /bin/login joedoe.

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Task 3 - Delete the User Account

For the final task, I needed to delete the joedoe account. I opened the Terminal and typed sudo userdel -r joedoe to delete the account along with its home directory. After confirming the deletion, I checked the /etc/passwd file using sudo less /etc/passwd to ensure the joedoe account was removed.

Finally, I cleared the Terminal by typing clear to prepare the workspace for any further tasks. This completed the process of creating, modifying, and deleting a user account in Ubuntu.

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Exercise 7 - Create and Configure Groups in Linux

Task 1 - Create Groups

In this task, I first ensured that I was connected to PLABCLIENT01 and had the Terminal window open. To create three new groups, I entered the commands sudo groupadd Marketing, sudo groupadd Sales, and sudo groupadd IT one after another in the Terminal. After executing these commands, I verified that the groups were successfully created by typing sudo tail /etc/group to view the contents of the /etc/group file and check for the presence of the new groups.

Task 2 - Add Users to Groups

Next, I moved on to creating new users and assigning them to the groups I had just created. I added three new users using the commands sudo useradd jimdoe, sudo useradd juliedoe, and sudo useradd jackdoe. With these users created, I then assigned them to the appropriate groups. I used the gpasswd command to add jimdoe to the Marketing group, juliedoe to the Sales group, and jackdoe to the IT group. To make juliedoe an admin of the Sales group, I ran sudo gpasswd -A juliedoe Sales. Additionally, to remove jimdoe from the Marketing group, I executed sudo gpasswd -d jimdoe Marketing.

Task 3 - Test Group Functionality

I then tested the functionality of the groups by creating files and observing their group ownership. I navigated to the Documents folder and created a new file named file1.txt using the touch command. To check the file's details, including its group ownership, I used ls -l. I then changed the active group to adm by using the newgrp adm command and created another file named file2.txt. Again, I checked the file's details with ls -l to confirm that the group for this file had changed to adm. To verify the users and groups, I ran getent group, which displayed the current group information and the users associated with each group. Lastly, I checked the details of the user juliedoe by executing getent passwd juliedoe. After completing these steps, I closed the Terminal window.

Jonathan Hernandez Rodriguez

Exercise 8 - Configure MFA in Linux

Task 1 - Update Ubuntu and Install the Google Authenticator for MFA

First, I connected to the PLABUBUNTU server and opened the Terminal by selecting the Show applications icon. In the Terminal, I initiated the update and upgrade process by typing sudo apt update && sudo apt upgrade and entering the password when prompted. After confirming the update by typing "Y" at the prompt, I waited for the upgrade to complete. Next, I installed the Google Authenticator package using the command sudo apt install libpam-google-authenticator. Once again, I entered the password and confirmed the installation by typing "Y". This ensured that the server was up-to-date and had the Google Authenticator installed.

Task 2 - Configure SSH to Work with the Google Authenticator

With the Google Authenticator installed, I proceeded to configure SSH to use it for MFA. I opened the SSH PAM configuration file by typing sudoedit /etc/pam.d/sshd in the Terminal. Inside the file, I commented out the @include common-auth line by adding a # at the beginning and then added the line auth required pam_google_authenticator.so immediately below it. After saving and exiting the file, I edited the SSH daemon configuration file by typing sudoedit /etc/ssh/sshd_config. In this file, I changed the ChallengeResponseAuthentication option from no to yes and added the following lines: PasswordAuthentication no, PubkeyAuthentication yes, and AuthenticationMethods publickey,keyboard-interactive. These changes were saved, and the file was closed.

Finally, I restarted the SSH service by typing sudo systemctl restart ssh to apply the changes. To clear the Terminal screen, I used the clear command and then minimized the Terminal window. Although not tested in this lab setting, the steps I followed enabled the use of Google Authenticator with Ubuntu Linux logins as part of an MFA setup, requiring both a password and the Google Authenticator for authentication.

Jonathan Hernandez Rodriguez

Task 3 - Configure Google Authenticator for a User

In this task, I configured Google Authenticator for a specific user account on the Ubuntu server. I began by opening the Terminal and ensuring I was logged in as the user who would use Google Authenticator. In the Terminal, I typed the command google-authenticator to start the configuration process. The tool generated a QR code and a secret key, which I scanned using the Google Authenticator app on my smartphone. I was then prompted to answer a series of questions, where I chose to enable time-based tokens, allow multiple uses of the same token, and set up rate-limiting to prevent brute-force attacks. After confirming the configuration, Google Authenticator was successfully set up for the user, adding an additional layer of security to the account by requiring a time-based one-time password (TOTP) during login.

Task 4 - Test SSH Login with MFA

After configuring Google Authenticator, I proceeded to test the SSH login process to ensure that MFA was working correctly. I opened the Terminal on a different machine and attempted to connect to the PLABUBUNTU server using SSH by typing ssh username@server_ip_address. As expected, I was first prompted to enter the user's password. After entering the password, I was then prompted to enter the TOTP generated by the Google Authenticator app on my smartphone. Upon entering the correct TOTP, I successfully logged into the server, verifying that the SSH configuration was correctly set up to require both a password and the Google Authenticator code for login. This task confirmed that the multi-factor authentication setup was functioning as intended, enhancing the security of remote access to the server.

Jonathan Hernandez Rodriguez

Exercise 9 - Configure Linux system to Join Microsoft Active Directory Domain for Authentication

Task 1 - Access and Update the Ubuntu System

I begin by connecting to the Ubuntu device. I open the applications menu by clicking on "Show Applications" in the bottom left corner and then scroll down to select the Terminal icon. Once the Terminal is open, I type sudo su to gain root access. If prompted, I enter the password Passw0rd. Next, I update the package list by typing apt update and pressing Enter. This command might take a few minutes to execute. Once complete, my system will be ready for installing the necessary packages.

Task 2 - Install Linux Domain Joining Tools

With the package list updated, I proceed by installing the domain joining tools. In the Terminal, I type apt install sssd-tools realmd adcli and press Enter. When the system prompts me with "Do you want to continue [Y/n]," I type Y and press Enter to confirm the installation of the required modules. I keep the Terminal window open after the installation is complete, as it will be needed for the next steps.

Task 3 - Configure Network Settings

Next, I access the network settings of the Ubuntu device. I click the "Show Applications" icon in the bottom left corner again, scroll down, and select "Settings." In the Settings window, I click on "Network" from the left-hand pane to open the network settings. Within the network settings, I locate the "Wired" section and click on the gear icon. This opens the Wired settings window. I switch to the IPv4 tab, where I modify the DNS settings. I slide the DNS slider from Automatic to Off, then enter 192.168.0.1, 192.168.0.6 in the provided textbox. I click "Apply" to save these settings.

Task 4 - Verify Network Connectivity

To apply the changes to the network settings, I return to the Terminal and restart the network manager by typing systemctl restart network-manager.service and pressing Enter. This command restarts the network manager service, ensuring that the new DNS settings are in effect. To verify that my Ubuntu device can reach the Active Directory domain, I type ping practicelabs.com into the Terminal and press Enter. The system starts pinging the domain to check connectivity. After a few seconds, I stop the ping command by pressing CTRL+C. A successful ping indicates that my device can reach the practicelabs.com domain.

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Task 5 - Domain Join the Ubuntu Device

I start by discovering the Active Directory domain. In the Terminal, I type realm discover -v practicelabs.com and press Enter. This command searches for the practicelabs.com domain and verifies its discoverability. If the domain is successfully discovered, I proceed to join the device to the domain. To join the domain, I type sudo realm -v join -U administrator plabdc01.practicelabs.com and press Enter in the Terminal. When prompted, I enter the password Passw0rd and press Enter. The system then connects my Ubuntu device to the practicelabs.com domain. Once the device is successfully joined, I can close the Terminal window. Finally, to verify that my Ubuntu device is now part of the domain, I connect to the Windows Server (PLABDC01). I open the Server Manager, click on "Tools," and select "Active Directory Users and Computers." In the Active Directory Users and Computers window, I select "Computers" from the left pane to see a list of devices. I locate the "Administrator-V" entry in the right pane, right-click on it, and select "Properties." This opens the Administrator-V Properties window. I navigate to the "Operating System" tab to verify that the device’s operating system is identified as Linux. This confirms that my Ubuntu device has been successfully joined to the Microsoft Active Directory domain.

Jonathan Hernandez Rodriguez

Assignment

Implementing Mandatory Access Control (MAC) on One World Bank's Linux Machines

When considering Mandatory Access Control (MAC) for One World Bank's Linux machines, it is essential to evaluate the strengths and weaknesses of three prominent options: SELinux, AppArmor, and local host permissions. Each of these approaches offers distinct advantages and challenges that can influence the overall security strategy.

SELinux (Security-Enhanced Linux) is a powerful and highly detailed MAC system.

It provides administrators with the ability to define specific permissions for processes and files through intricate policies. This granular control is a significant advantage because it enhances security by precisely regulating how processes interact with one another and with system resources. For instance, SELinux can enforce strict policies on which processes can access specific files, thereby preventing unauthorized access and potential data breaches. However, the complexity of SELinux can pose challenges, particularly for those who are not well-versed in its configuration. Crafting and managing SELinux policies requires a deep understanding of the system’s operations and potential security threats. Additionally, the extensive checks involved in SELinux's detailed policy management can introduce some performance overhead, potentially affecting system efficiency.

AppArmor (Application Armor) offers a simpler and more user-friendly alternative to SELinux. It utilizes profile-based controls that are generally easier to configure and manage.

This simplicity is particularly beneficial for environments where quick deployment and straightforward management are priorities. AppArmor profiles allow administrators to specify what resources an application can access, providing a level of security that is less complex to implement compared to SELinux. Moreover, AppArmor usually incurs less performance overhead, making it a good choice for systems where resource efficiency is a concern. However, the simplicity of AppArmor also means it offers less granular control compared to SELinux. While AppArmor is effective in managing application permissions, it provides fewer details and less flexibility in addressing complex security scenarios. This limitation can be a drawback in environments with sophisticated security requirements or where precise control over system interactions is necessary.

Local Host Permissions involve setting file and directory permissions directly on the system. This approach is straightforward and allows for immediate effects, making it relatively easy to implement and manage. Administrators can quickly adjust permissions to restrict or grant access to specific files and directories, providing a basic level of security. However, local host permissions offer a limited scope of control. They are primarily concerned with managing access to individual files and directories, which means they do not address inter-process communications or system-wide security comprehensively. Additionally, managing permissions across multiple machines can become cumbersome and inconsistent, especially in larger or more complex environments. This lack of centralized control can lead to security gaps and administrative challenges.

For One World Bank's Linux machines, SELinux is recommended despite its inherent complexity. The primary reason for this recommendation is SELinux's robust security features and its capacity to offer detailed, granular control over system resources. In a high-stakes environment such as a bank, where protecting sensitive financial data and ensuring compliance with stringent security standards are critical, SELinux provides the comprehensive access controls necessary to mitigate potential threats effectively. While SELinux requires a more sophisticated understanding and may introduce some performance overhead, its ability to deliver extensive and precise security measures outweighs these drawbacks. By implementing SELinux, One World Bank can enhance its security posture, safeguard sensitive information, and maintain a high level of protection against various security threats.

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